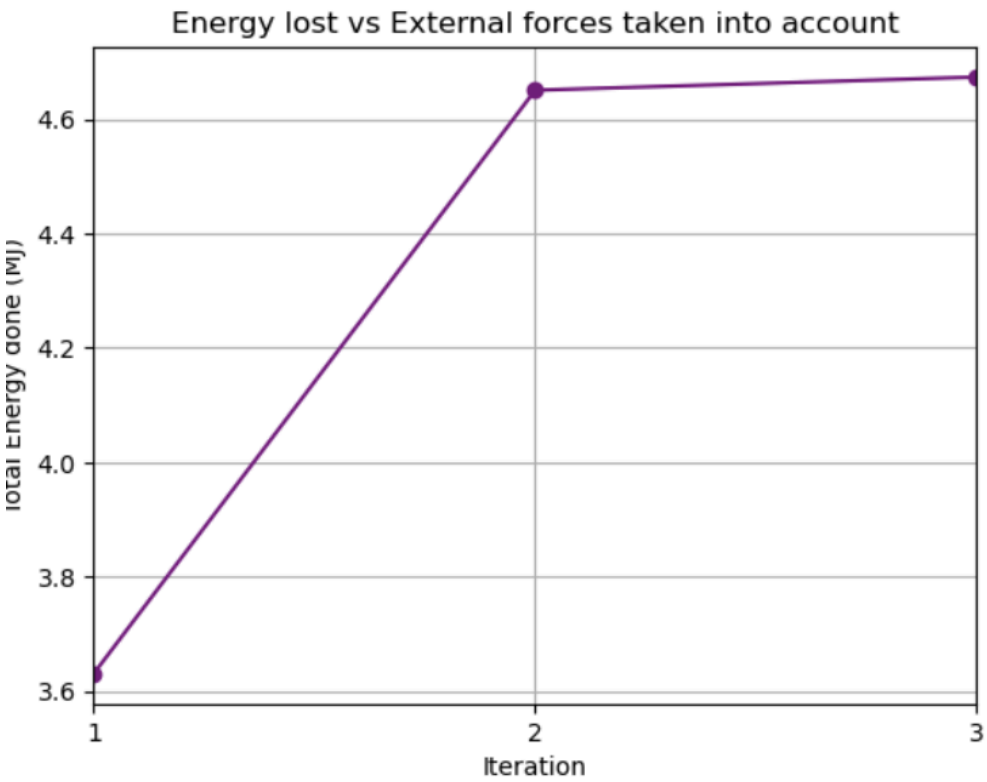
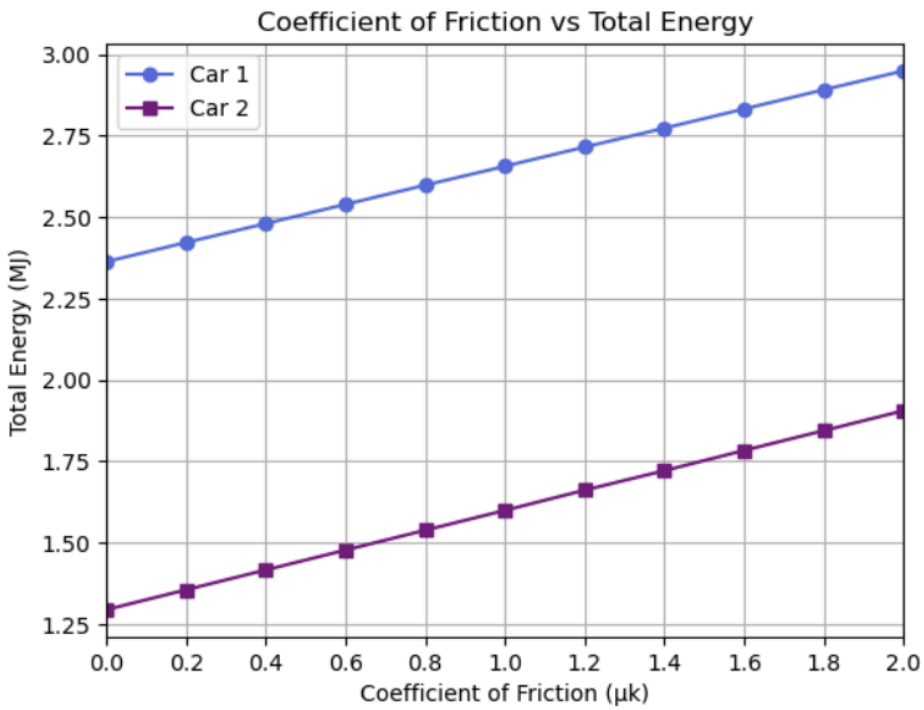
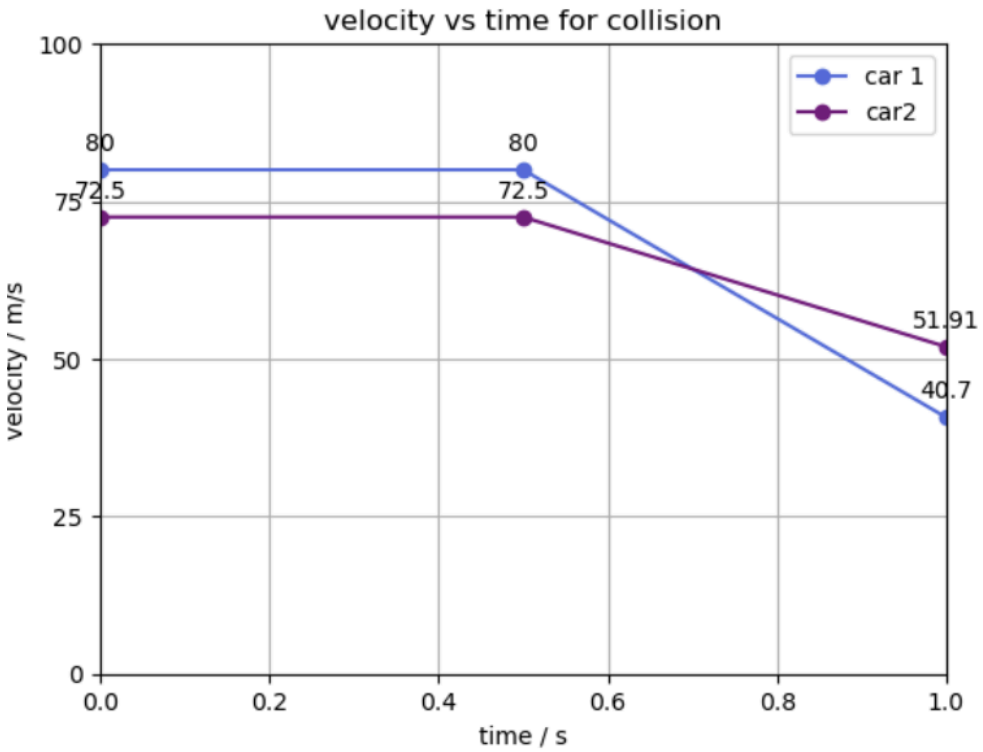
```python
import matplotlib.pyplot as plt
import numpy as np

# Data
iterations = [1, 2, 3]
energy_mj = [3.63, 4.651, 4.674]

plt.figure()

# Plot line
plt.plot(iterations, energy_mj, marker='o', color='purple')

#Set x limit
plt.xlim(1, 3)

# Labels and title
plt.title("Energy lost vs External forces taken into account")
plt.xlabel("Iteration")
plt.ylabel("Total Energy done (MJ)")

# Set x-axis ticks every 1
plt.xticks(np.arange(1, 4, 1))

# Grid
plt.grid(True)

plt.show()
```



```python
import matplotlib.pyplot as plt

# Coefficient of friction values
mu = [0, 0.2, 0.4, 0.6, 0.8, 1, 1.2, 1.4, 1.6, 1.8, 2]

# Total work (MJ)
car1_work = [2.363, 2.422, 2.480, 2.539, 2.598, 2.656, 2.715, 2.773, 2.832, 2.891, 2.949]
car2_work = [1.294, 1.355, 1.416, 1.477, 1.538, 1.599, 1.661, 1.721, 1.783, 1.844, 1.905]

# Plot
plt.plot(mu, car1_work, marker='o', color='royalblue', label="Car 1")
plt.plot(mu, car2_work, marker='s', color='purple', label="Car 2")

#Force x-axis to start exactly at 0
plt.xlim(0,2)
plt.xticks([0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0])

# Labels and title
plt.xlabel("Coefficient of Friction (μk)")
plt.ylabel("Total Energy (MJ)")
plt.title("Coefficient of Friction vs Total Energy")
plt.legend()
plt.grid(True)

# Show graph
plt.show()
```



```python
import matplotlib.pyplot as plt

# Time values
time = [0, 0.5, 1]

# Velocities
car1 = [80, 80, 40.7]
car2 = [72.5, 72.5, 51.91]

# Plot lines
plt.plot(time, car1, marker='o', color='royalblue', label='car 1')
plt.plot(time, car2, marker='o', color='purple', label='car2')

# Title and labels
plt.title("velocity vs time for collision")
plt.xlabel("time / s")
plt.ylabel("velocity / m/s")

#Force x-axis to start at exactly 0
plt.xlim(0,1)

# Show full y-axis range
plt.ylim(0, 100)
plt.yticks(range(0, 101, 25))  # 0,25,50,75,100

#Show x-axis ticks every 0.5
plt.xlim(0,1)  #0,50,100

# Value labels
for x, y in zip(time, car1):
    plt.text(x, y + 3, str(y), ha='center')

for x, y in zip(time, car2):
    plt.text(x, y + 3, str(y), ha='center')

# Grid and legend
plt.grid(True)
plt.legend()

plt.show()
```